

SER-IW and the Shared- ω Approximation

1 Goal

The purpose of this note is to separate two objects:

1. the more principled single-effect inverse-Wishart model, called SER-IW here;
2. a tractable approximation that leads to a fast shared- ω SER variant.

The shared- ω approximation is not designed to make the SER PIP exactly ideal. Instead, the goal is to make the approximation explicit, fast, and amenable to later bounding of the SNP-specific PIP distortion.

2 SER-IW Model

Let $x \in \mathbb{R}^J$ be the vector of marginal summary statistics, $\bar{R}$ be the reference LD matrix, and N be the GWAS sample size. The single-effect inverse-Wishart model is

$$R \sim \mathcal{IW}(\nu_0 \bar{R}, \nu_0 + J + 1), \quad \gamma \sim \text{Unif}\{1, \dots, J\}, \quad \beta \sim \mathcal{N}(0, \sigma_0^2),$$

and

$$x \mid R, \beta, \gamma = j \sim \mathcal{N}(\beta R_{\cdot j}, R/N).$$

This is the original natural model in which the same unknown in-sample LD matrix R appears in both the mean and the noise covariance.

2.1 Block decomposition

For a fixed active SNP j , write

$$d_j = R_{jj}, \quad z_j = \frac{R_{-j,j}}{d_j}, \quad S_j = R_{-j,-j} - d_j z_j z_j^\top.$$

Then

$$R = \begin{pmatrix} d_j & d_j z_j^\top \\ d_j z_j & S_j + d_j z_j z_j^\top \end{pmatrix}.$$

Let

$$\mu_j = \bar{R}_{-j,j}, \quad \bar{S}_j = \bar{R}_{-j,-j} - \mu_j \mu_j^\top.$$

The inverse-Wishart block identities give the working representation

$$d_j \sim \text{IG}\left(\frac{\nu_0 + 2}{2}, \frac{\nu_0}{2}\right), \quad z_j \mid S_j \sim \mathcal{N}\left(\mu_j, \frac{S_j}{\nu_0}\right),$$

and

$$S_j \sim \mathcal{IW}(\nu_0 \bar{S}_j, \nu_0 + J + 1),$$

using the same convention as in the earlier notes.

2.2 Useful likelihood factorization

Under the SER-IW likelihood,

$$x_j \mid d_j, \beta, \gamma = j \sim \mathcal{N}(\beta d_j, d_j/N),$$

and the Gaussian conditioning formula gives

$$x_{-j} \mid x_j, z_j, S_j, \gamma = j \sim \mathcal{N}(x_j z_j, S_j/N).$$

The important point is that the conditional distribution of x_{-j} given x_j , z_j , and S_j does not involve β or d_j .

If we fix $d_j \equiv 1$, as in the model-B simplification, this becomes

$$x_j \mid \beta, \gamma = j \sim \mathcal{N}(\beta, 1/N), \quad x_{-j} \mid x_j, z_j, S_j, \gamma = j \sim \mathcal{N}(x_j z_j, S_j/N).$$

This factorization is the source of the ideal-PIP behavior in the single-effect model: the active coordinate carries the β -evidence, whereas the nonactive coordinates learn the LD column.

2.3 Exact conditional marginal under the $d_j = 1$ simplification

Conditioning on S_j , the prior and likelihood for z_j are conjugate:

$$z_j \mid S_j \sim \mathcal{N}(\mu_j, S_j/\nu_0), \quad x_{-j} \mid x_j, z_j, S_j \sim \mathcal{N}(x_j z_j, S_j/N).$$

Integrating over z_j gives

$$x_{-j} \mid x_j, S_j, \gamma = j \sim \mathcal{N}\left(x_j \mu_j, \left(\frac{1}{N} + \frac{x_j^2}{\nu_0}\right) S_j\right).$$

After integrating over S_j , this becomes a multivariate t -type conditional distribution centered at $x_j \mu_j$. Equivalently, it can be represented using a normal–Gamma mixture:

$$x_{-j} \mid x_j, \omega_j, \gamma = j \sim \mathcal{N}\left(x_j \mu_j, \frac{1}{\omega_j} \left(\frac{1}{N} + \frac{x_j^2}{\nu_0}\right) \bar{S}_j\right),$$

$$\omega_j \sim \text{Gamma}\left(\frac{\nu_0 + 3}{2}, \frac{\nu_0}{2}\right),$$

up to the precise scale convention used for the multivariate t . This conditional representation is close to ideal, but it is awkward to extend to IBSS because the conditional covariance depends on the observed active coordinate x_j and only describes $x_{-j} \mid x_j$, not a common Gaussian residual model for the whole vector.

3 Approximation Leading to Shared- ω

The shared- ω approximation replaces the exact conditional matrix-valued uncertainty by a scalar local scale. For each candidate active SNP j , introduce ω_j and set

$$\omega_j \sim \text{Gamma}(a_\omega, b_\omega), \quad a_\omega = \frac{\nu_0 + 3}{2}, \quad b_\omega = \frac{\nu_0}{2}.$$

The same ω_j scales both the observation noise and the relaxed-column prior:

$$x \mid \beta, \gamma = j, z_j, \omega_j \sim \mathcal{N}_J \left(\beta u_j(z_j), \frac{\bar{R}}{N\omega_j} \right),$$

$$z_j \mid \omega_j \sim \mathcal{N}_{J-1} \left(\mu_j, \frac{\bar{S}_j}{\nu_0 \omega_j} \right).$$

Here $u_j(z_j)$ is the J -vector whose j -th entry is one and whose remaining entries are z_j .

This approximation makes three deliberate simplifications.

1. The random Schur complement S_j is replaced by its reference value $\bar{S}_j$, up to the scalar scale $1/\omega_j$.
2. The same scalar ω_j controls uncertainty in both the noise and the LD column. This is what keeps the posterior update for ω_j conjugate.
3. The conditional model $x_{-j} \mid x_j$ is replaced by a full-vector Gaussian likelihood with covariance $\bar{R}/(N\omega_j)$. This is less exact for SER PIPs, but much easier to embed in an IBSS residual algorithm.

My attempt to justify the approximation

Write the model as

$$x = \beta \begin{pmatrix} 1 \\ z_1 \end{pmatrix} + z_0, \tag{1}$$

where

$$\begin{cases} z_1 \mid S_1 \sim \mathcal{N}(\mu_1, S_1/\nu_0) \\ z_0 \sim \mathcal{N}(0, R/N) \\ S_1 \text{ is the Schur's complement of } R_{11} \text{ in } R \end{cases} \tag{2}$$

Note that this induces:

$$S_1 \sim \mathcal{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1).$$

Our goal is to show that this generative distribution of (z_0, z_1) approximates:

$$\begin{cases} z_1 \mid \omega \sim \mathcal{N} \left(\mu_1, \frac{\bar{S}_1}{\omega \nu_0} \right) \\ z_0 \mid \omega \sim \mathcal{N} \left(0, \frac{\bar{R}}{\omega N} \right) \\ \omega \sim \text{Gamma}((\nu_0 + 2)/2, \nu_0/2). \end{cases} \tag{3}$$

Approach 0

We can easily show that the marginal distribution of z_0 is the same in both models. Can we show that $p(z_1 | z_0)$ from (2) approximates $\tilde{p}(z_1 | z_0)$ from (3)?

Write

$$z_0 = \begin{pmatrix} z_{01} \\ z_{0,-1} \end{pmatrix}, \quad \delta = z_{0,-1} - \mu_1 z_{01}, \quad \kappa = \nu_0 + N z_{01}^2.$$

Under the inverse-Wishart model,

$$R | z_0 \sim \mathcal{IW}(\nu_0 \bar{R} + N z_0 z_0^\top, \nu_0 + J + 2).$$

Using the block inverse-Wishart identities,

$$z_1 | S_1, z_0 \sim \mathcal{N}\left(\mu_1 + \frac{N z_{01}}{\kappa} \delta, \frac{S_1}{\kappa}\right),$$

and

$$S_1 | z_0 \sim \mathcal{IW}\left(\nu_0 \bar{S}_1 + \frac{\nu_0 N}{\kappa} \delta \delta^\top, \nu_0 + J + 2\right).$$

Thus the exact conditional distribution $p(z_1 | z_0)$ is centered at

$$\mu_1 + \frac{N z_{01}}{\nu_0 + N z_{01}^2} (z_{0,-1} - \mu_1 z_{01}),$$

not at μ_1 . It also contains a rank-one update to the Schur complement scale through

$$\frac{\nu_0 N}{\kappa} \delta \delta^\top.$$

In the shared- ω approximation,

$$z_0 | \omega \sim \mathcal{N}\left(0, \frac{\bar{R}}{N\omega}\right), \quad z_1 | \omega \sim \mathcal{N}\left(\mu_1, \frac{\bar{S}_1}{\nu_0 \omega}\right),$$

and z_0 and z_1 are conditionally independent given ω . Conditioning on z_0 updates ω , but it does not shift the conditional mean of z_1 . Therefore

$$\mathbb{E}_{\text{shared-}\omega}(z_1 | z_0) = \mu_1.$$

This means that the shared- ω approximation does not generally match the exact conditional distribution $p(z_1 | z_0)$. It ignores the z_0 -dependent mean shift

$$\frac{N z_{01}}{\nu_0 + N z_{01}^2} (z_{0,-1} - \mu_1 z_{01})$$

and the rank-one covariance update

$$\frac{\nu_0 N}{\nu_0 + N z_{01}^2} (z_{0,-1} - \mu_1 z_{01})(z_{0,-1} - \mu_1 z_{01})^\top.$$

The approximation is most plausible when this conditional residual effect is small, for example when ν_0 is large, z_{01} is small, or $z_{0,-1} - \mu_1 z_{01}$ is small. This is a useful baseline comparison to return to later if we want to bound the approximation error more carefully.

3.1 Strategy 1

Strategy 1 starts from the normal–Gamma representation of the marginal prior for the LD column:

$$z_1 \mid \omega \sim \mathcal{N}_{J-1} \left(\mu_1, \frac{\bar{S}_1}{\nu_0 \omega} \right), \quad \omega \sim \text{Gamma} \left(\frac{\nu_0 + 3}{2}, \frac{\nu_0}{2} \right).$$

The question is whether, after introducing this ω , the noise vector z_0 can be approximated by

$$z_0 \mid \omega \approx \mathcal{N}_J \left(0, \frac{\bar{R}}{N\omega} \right).$$

For clarity, work with the $d_1 = 1$ version. Conditional on the LD column z_1 and Schur complement S_1 , the covariance matrix is

$$R = \begin{pmatrix} 1 & z_1^\top \\ z_1 & S_1 + z_1 z_1^\top \end{pmatrix}.$$

Thus

$$z_0 \mid z_1, S_1 \sim \mathcal{N}_J \left(0, \frac{1}{N} \begin{pmatrix} 1 & z_1^\top \\ z_1 & S_1 + z_1 z_1^\top \end{pmatrix} \right).$$

Equivalently,

$$z_{01} \sim \mathcal{N}(0, 1/N), \quad z_{0,-1} \mid z_{01}, z_1, S_1 \sim \mathcal{N}_{J-1} \left(z_{01} z_1, \frac{S_1}{N} \right).$$

The column mixture above can be viewed as replacing the random Schur complement by the scalar approximation

$$S_1 \approx \frac{\bar{S}_1}{\omega}.$$

Under this approximation,

$$z_0 \mid z_1, \omega \approx \mathcal{N}_J \left(0, \frac{1}{N} \begin{pmatrix} 1 & z_1^\top \\ z_1 & z_1 z_1^\top + \bar{S}_1/\omega \end{pmatrix} \right).$$

Equivalently,

$$z_{01} \sim \mathcal{N}(0, 1/N), \quad z_{0,-1} \mid z_{01}, z_1, \omega \sim \mathcal{N}_{J-1} \left(z_{01} z_1, \frac{\bar{S}_1}{N\omega} \right).$$

This is the direct consequence of Strategy 1. It is not the same as

$$\mathcal{N}_J \left(0, \frac{\bar{R}}{N\omega} \right), \quad \bar{R} = \begin{pmatrix} 1 & \mu_1^\top \\ \mu_1 & \bar{S}_1 + \mu_1 \mu_1^\top \end{pmatrix}.$$

The direct Strategy 1 covariance is

$$\frac{1}{N} \begin{pmatrix} 1 & z_1^\top \\ z_1 & z_1 z_1^\top + \bar{S}_1/\omega \end{pmatrix},$$

whereas the desired shared- ω covariance is

$$\frac{1}{N\omega} \begin{pmatrix} 1 & \mu_1^\top \\ \mu_1 & \bar{S}_1 + \mu_1 \mu_1^\top \end{pmatrix}.$$

Therefore, starting from the z_1 mixture does not by itself justify scaling the entire noise covariance by $1/\omega$. It justifies scaling the conditional Schur-complement noise S_1/N by $1/\omega$.

To reach the desired shared- ω likelihood, we would need additional approximations, for example:

1. replace z_1 by its center μ_1 in the noise covariance;
2. also scale the active-coordinate and rank-one parts of the covariance by $1/\omega$.

The first approximation is natural when LD-column uncertainty is small. The second is less directly implied by the column mixture. It may still be useful computationally, but it should be treated as an extra approximation rather than a consequence of the SER-IW block decomposition.

This suggests an alternative Strategy 1 approximation:

$$z_0 \mid z_1, \omega \approx \mathcal{N}_J \left(0, \frac{1}{N} \begin{pmatrix} 1 & z_1^\top \\ z_1 & z_1 z_1^\top + \bar{S}_1/\omega \end{pmatrix} \right),$$

or a stabilized version replacing z_1 by μ_1 only in the covariance. This is more faithful to the column-first derivation, but it is less convenient than the current shared- ω model because the likelihood covariance depends on the active column.

3.2 Strategy 2

Strategy 2 starts from the normal-Gamma representation of the marginal noise vector:

$$z_0 \mid \omega \sim \mathcal{N}_J \left(0, \frac{\bar{R}}{N\omega} \right), \quad \omega \sim \text{Gamma} \left(\frac{\nu_0 + 2}{2}, \frac{\nu_0}{2} \right).$$

The question is whether, after introducing this ω , the LD column can be approximated by

$$z_1 \mid \omega \approx \mathcal{N}_{J-1} \left(\mu_1, \frac{\bar{S}_1}{\nu_0 \omega} \right).$$

Using the block form

$$\bar{R} = \begin{pmatrix} 1 & \mu_1^\top \\ \mu_1 & \bar{S}_1 + \mu_1 \mu_1^\top \end{pmatrix},$$

the noise mixture implies

$$z_{01} \mid \omega \sim \mathcal{N} \left(0, \frac{1}{N\omega} \right),$$

and

$$z_{0,-1} \mid z_{01}, \omega \sim \mathcal{N}_{J-1} \left(\mu_1 z_{01}, \frac{\bar{S}_1}{N\omega} \right).$$

Equivalently, the conditional noise residual

$$\delta_0 = z_{0,-1} - \mu_1 z_{01}$$

satisfies

$$\delta_0 \mid \omega \sim \mathcal{N}_{J-1} \left(0, \frac{\bar{S}_1}{N\omega} \right), \quad \delta_0 \perp z_{01} \mid \omega.$$

Now use the observation equation

$$x = \beta \begin{pmatrix} 1 \\ z_1 \end{pmatrix} + z_0.$$

For $\gamma = 1$,

$$x_1 = \beta + z_{01}, \quad x_{-1} = \beta z_1 + z_{0,-1}.$$

Subtracting $\mu_1 x_1$ gives

$$x_{-1} - \mu_1 x_1 = \beta(z_1 - \mu_1) + \delta_0.$$

Thus, conditional on ω ,

$$x_{-1} - \mu_1 x_1 \mid \beta, z_1, \omega \sim \mathcal{N}_{J-1} \left(\beta(z_1 - \mu_1), \frac{\bar{S}_1}{N\omega} \right).$$

If we now place the working prior

$$z_1 \mid \omega \sim \mathcal{N}_{J-1} \left(\mu_1, \frac{\bar{S}_1}{\nu_0 \omega} \right),$$

then the conditional residual has the marginal distribution

$$x_{-1} - \mu_1 x_1 \mid \beta, \omega \sim \mathcal{N}_{J-1} \left(0, \frac{1}{\omega} \left(\frac{1}{N} + \frac{\beta^2}{\nu_0} \right) \bar{S}_1 \right).$$

This is close in form to the exact model-B conditional distribution, except that x_1^2 is replaced by β^2 :

$$x_{-1} - \mu_1 x_1 \mid x_1 \quad \text{exactly has scale roughly} \quad \left(\frac{1}{N} + \frac{x_1^2}{\nu_0} \right) \bar{S}_1.$$

Therefore Strategy 2 gives a coherent full-vector Gaussian model, but it changes the conditioning logic: the nonactive residual is explained through $\beta(z_1 - \mu_1)$, rather than through $x_1(z_1 - \mu_1)$.

Compared with Strategy 1, Strategy 2 more naturally justifies scaling the entire observation-noise covariance by $1/\omega$, because this is how the strategy starts. The additional approximation is instead in the LD-column prior: we attach the same noise scale ω to $z_1 - \mu_1$, even though the exact inverse-Wishart dependence between z_0 and z_1 is more complicated.

This strategy leads directly to the shared- ω model:

$$z_0 \mid \omega \sim \mathcal{N}_J \left(0, \frac{\bar{R}}{N\omega} \right), \quad z_1 \mid \omega \sim \mathcal{N}_{J-1} \left(\mu_1, \frac{\bar{S}_1}{\nu_0 \omega} \right),$$

$$x = \beta \begin{pmatrix} 1 \\ z_1 \end{pmatrix} + z_0.$$

In this sense, Strategy 2 is a more direct justification for the current shared- ω likelihood, while Strategy 1 is a more direct justification for scaling the Schur-complement part of the LD uncertainty.

4 SER Variational Inference for Shared- ω

Use the mean-field approximation

$$q(\gamma = j, \beta, z_j, \omega_j) = \pi_j q_j(\beta) q_j(z_j) q_j(\omega_j),$$

with

$$q_j(\beta) = \mathcal{N}(m_{\beta j}, v_{\beta j}), \quad q_j(z_j) = \mathcal{N}(m_{zj}, V_{zj}), \quad q_j(\omega_j) = \text{Gamma}(\alpha_{\omega j}, \beta_{\omega j}).$$

Let

$$\bar{\omega}_j = \mathbb{E}_j(\omega_j), \quad \bar{\beta}_j = m_{\beta j}, \quad \bar{\beta}^2_j = v_{\beta j} + m_{\beta j}^2.$$

Define

$$\Omega = \bar{R}^{-1}, \quad C_j(x) = x^\top \Omega x - x_j^2.$$

For the current $q_j(z_j)$, define

$$\begin{aligned} Q_j &= \text{tr}(\bar{S}_j^{-1} V_{zj}) + (m_{zj} - \mu_j)^\top \bar{S}_j^{-1} (m_{zj} - \mu_j), \\ A_j &= 1 + Q_j, \\ B_j &= x_j + (m_{zj} - \mu_j)^\top \bar{S}_j^{-1} (x_{-j} - \mu_j x_j). \end{aligned}$$

The beta update is

$$v_{\beta j} = (\sigma_0^{-2} + N \bar{\omega}_j A_j)^{-1}, \quad m_{\beta j} = v_{\beta j} N \bar{\omega}_j B_j.$$

The relaxed-column update is

$$\begin{aligned} V_{zj} &= \frac{\bar{S}_j}{\bar{\omega}_j (N \bar{\beta}^2_j + \nu_0)}, \\ m_{zj} &= \mu_j + \frac{N \bar{\beta}_j}{N \bar{\beta}^2_j + \nu_0} (x_{-j} - \mu_j x_j). \end{aligned}$$

The Gamma update is

$$\begin{aligned} \alpha_{\omega j} &= a_\omega + \frac{J}{2} + \frac{J-1}{2}, \\ \beta_{\omega j} &= b_\omega + \frac{1}{2} \left[N \left\{ x^\top \Omega x - 2 \bar{\beta}_j B_j + \bar{\beta}^2_j A_j \right\} + \nu_0 Q_j \right]. \end{aligned}$$

The coordinate weights are

$$\pi_j = \frac{\exp(\mathcal{L}_j)}{\sum_{k=1}^J \exp(\mathcal{L}_k)},$$

where $\mathcal{L}_j$ is the local variational lower bound including the likelihood, the priors for β, z_j, ω_j , and the entropies of $q_j(\beta), q_j(z_j), q_j(\omega_j)$.

5 PIP Distortion View

The ideal SER weight has the form

$$\ell_j^{\text{ideal}} = \text{const} + \frac{N}{2} \frac{N \sigma_0^2}{1 + N \sigma_0^2} x_j^2.$$

The shared- ω approximation produces some local weight

$$\ell_j^{\text{sh-}\omega} = \ell_j^{\text{ideal}} + \delta_j.$$

The goal is not necessarily to force $\delta_j = 0$. Instead, the useful target is to understand and bound the spread

$$\Delta = \max_j \delta_j - \min_j \delta_j.$$

If Δ is small, then the approximate PIPs are uniformly close to the ideal PIPs in the sense that their odds are distorted by at most a factor $\exp(\Delta)$. This gives a concrete way to judge whether the shared- ω approximation is acceptable.